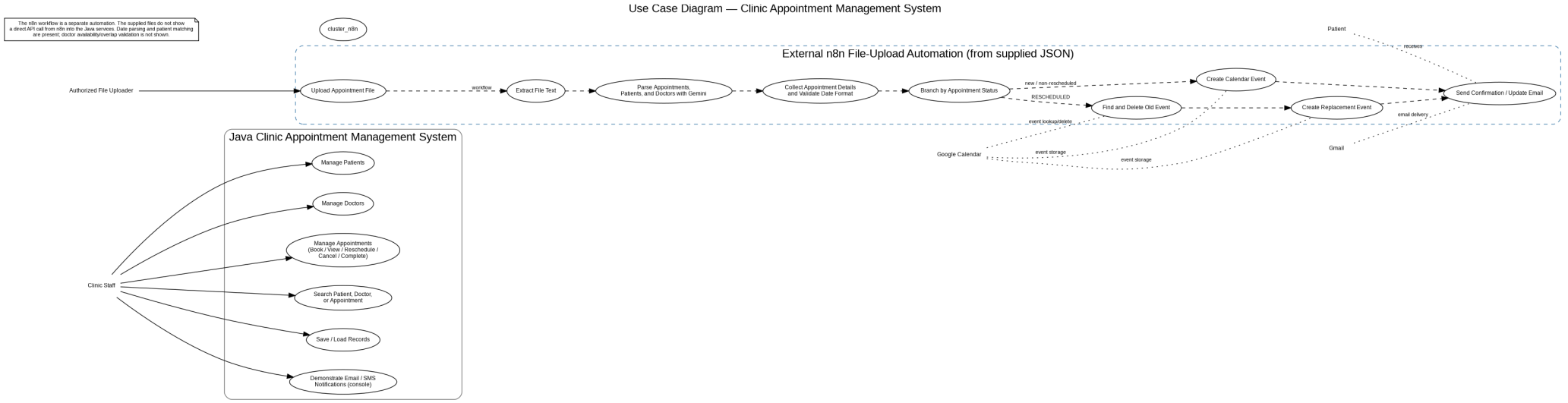
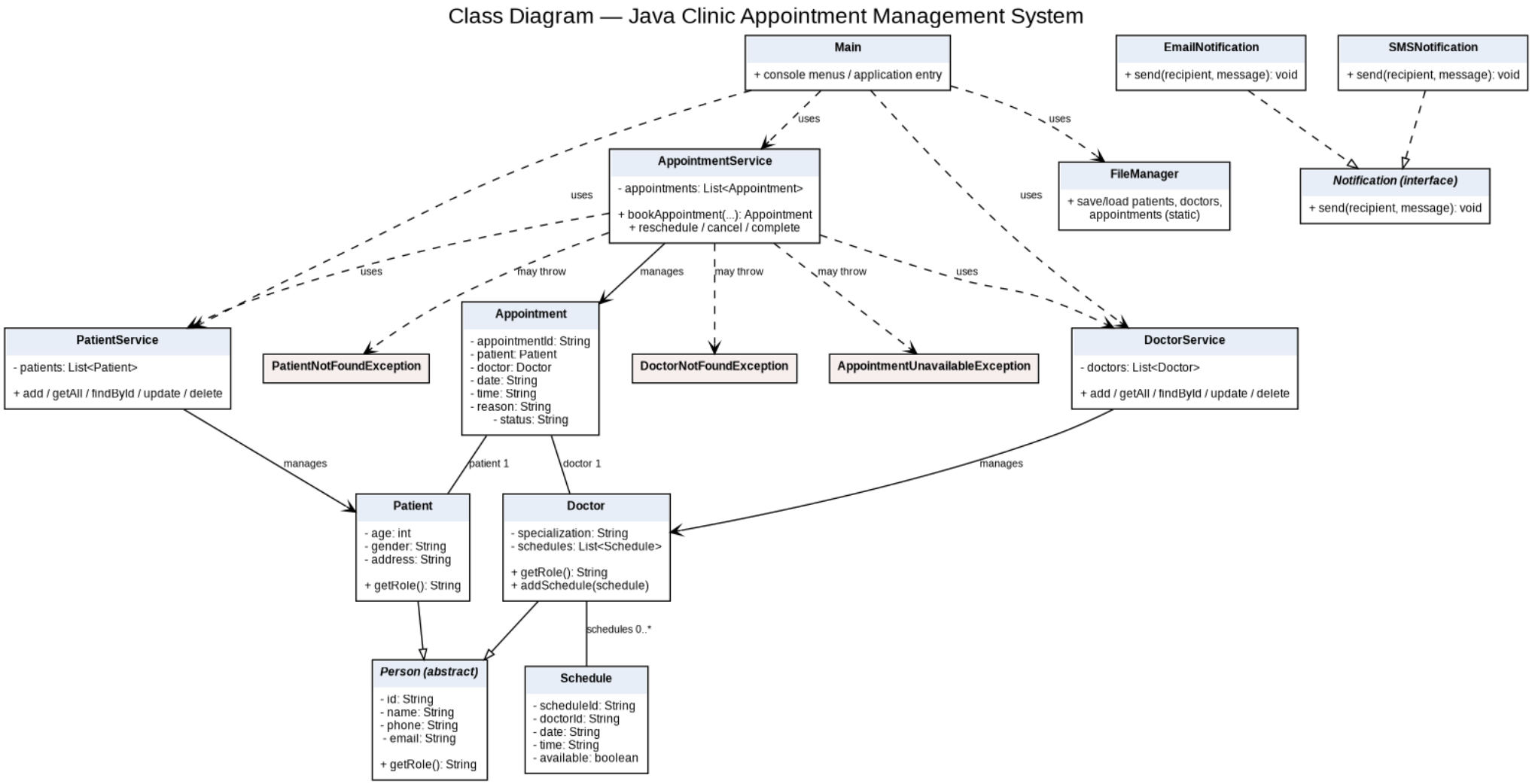
1. Use Case Diagram

Shows Java clinic-system functions separately from the n8n file-upload automation and its external Gemini, Google Calendar, and Gmail services.



2. Class Diagram

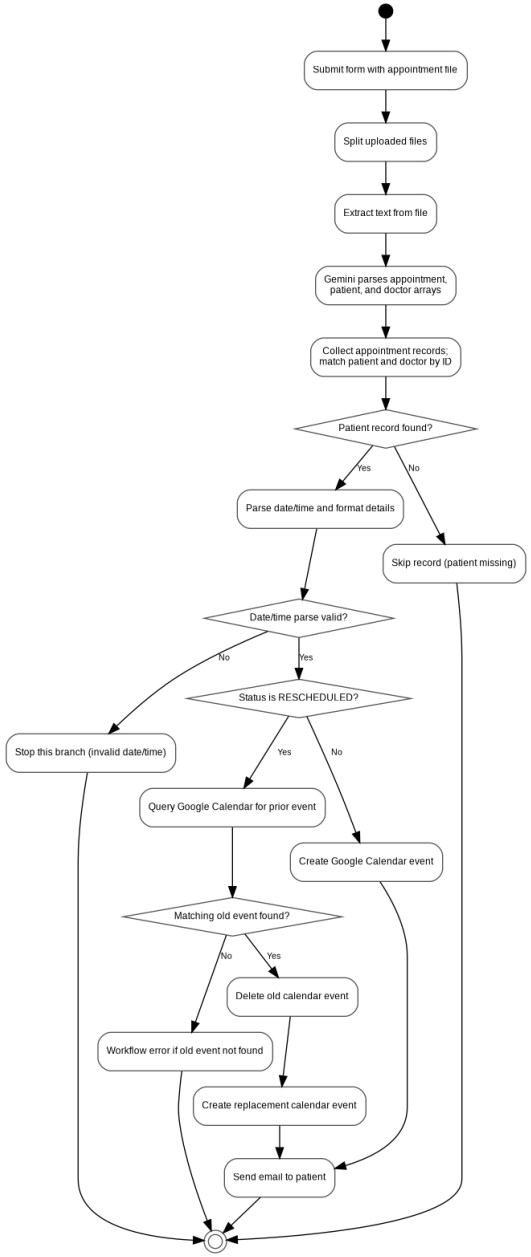
Models the Java classes, inheritance, Notification interface implementations, associations, and service dependencies from the provided source files.



3. Activity Diagram — Appointment File Upload Automation

Follows the n8n workflow: upload, extract, parse with Gemini, collect appointment records, validate patient/date information, branch by status, update Calendar, and email.

Activity Diagram — Appointment File Upload Automation



4. Sequence Diagram — File Upload to Calendar Booking

Shows the ordered interactions between the authorized user, n8n nodes, Gemini, Google Calendar, and Gmail, including the rescheduling path.

